

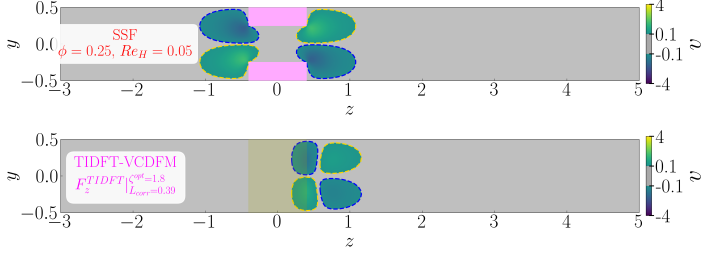
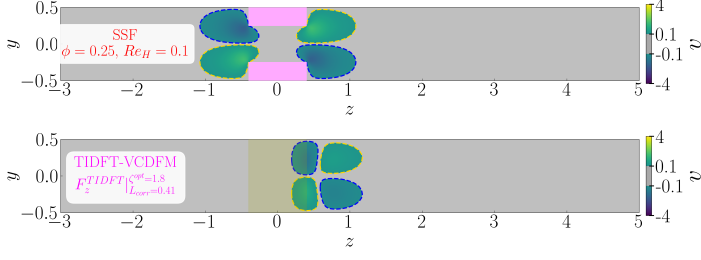
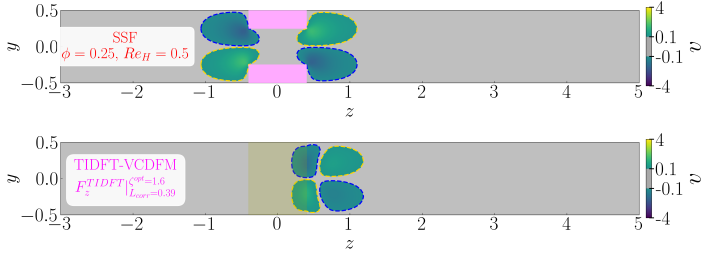
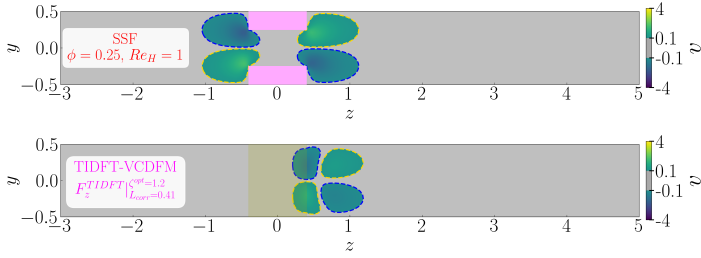
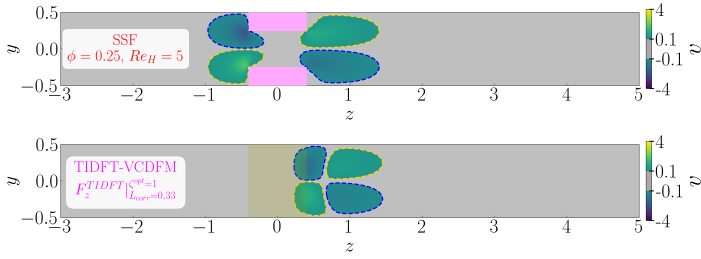
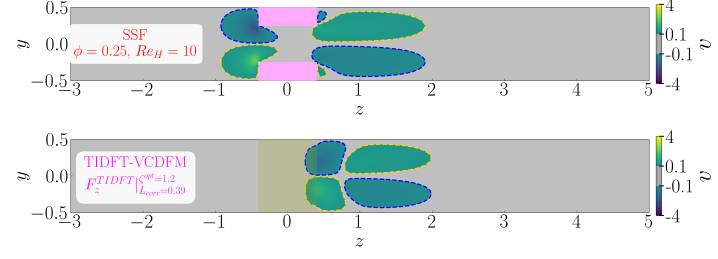
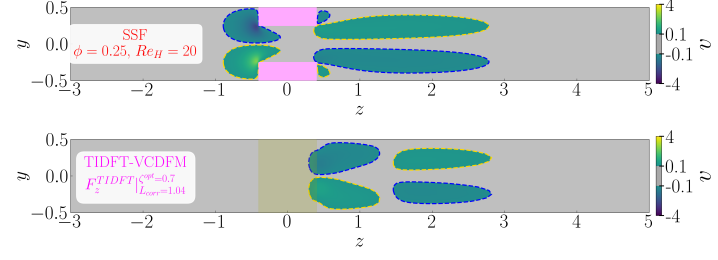
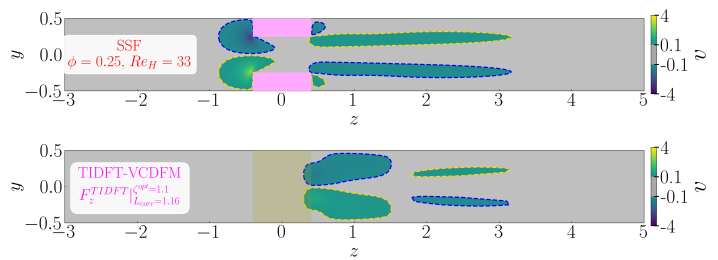
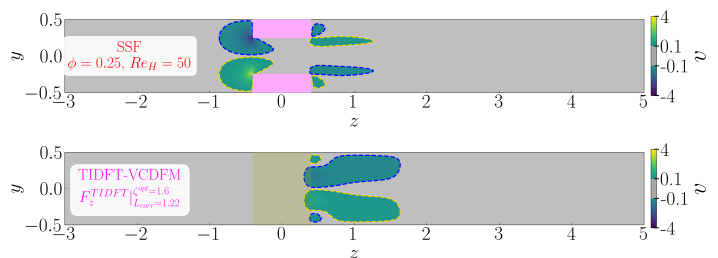
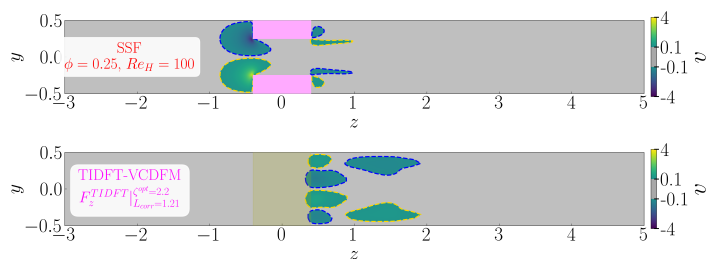
$\phi = 0.25$

(a) $Re_H = 0.05$

(b) $Re_H = 0.1$

(c) $Re_H = 0.5$

(d) $Re_H = 1$

(e) $Re_H = 5$

(f) $Re_H = 10$

(g) $Re_H = 20$

(h) $Re_H = 33.33$

(i) $Re_H = 50$

(j) $Re_H = 100$

Figure D.13: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.25$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the TIDFT-VCDFM reconstruction using F_z^{TIDFT} .

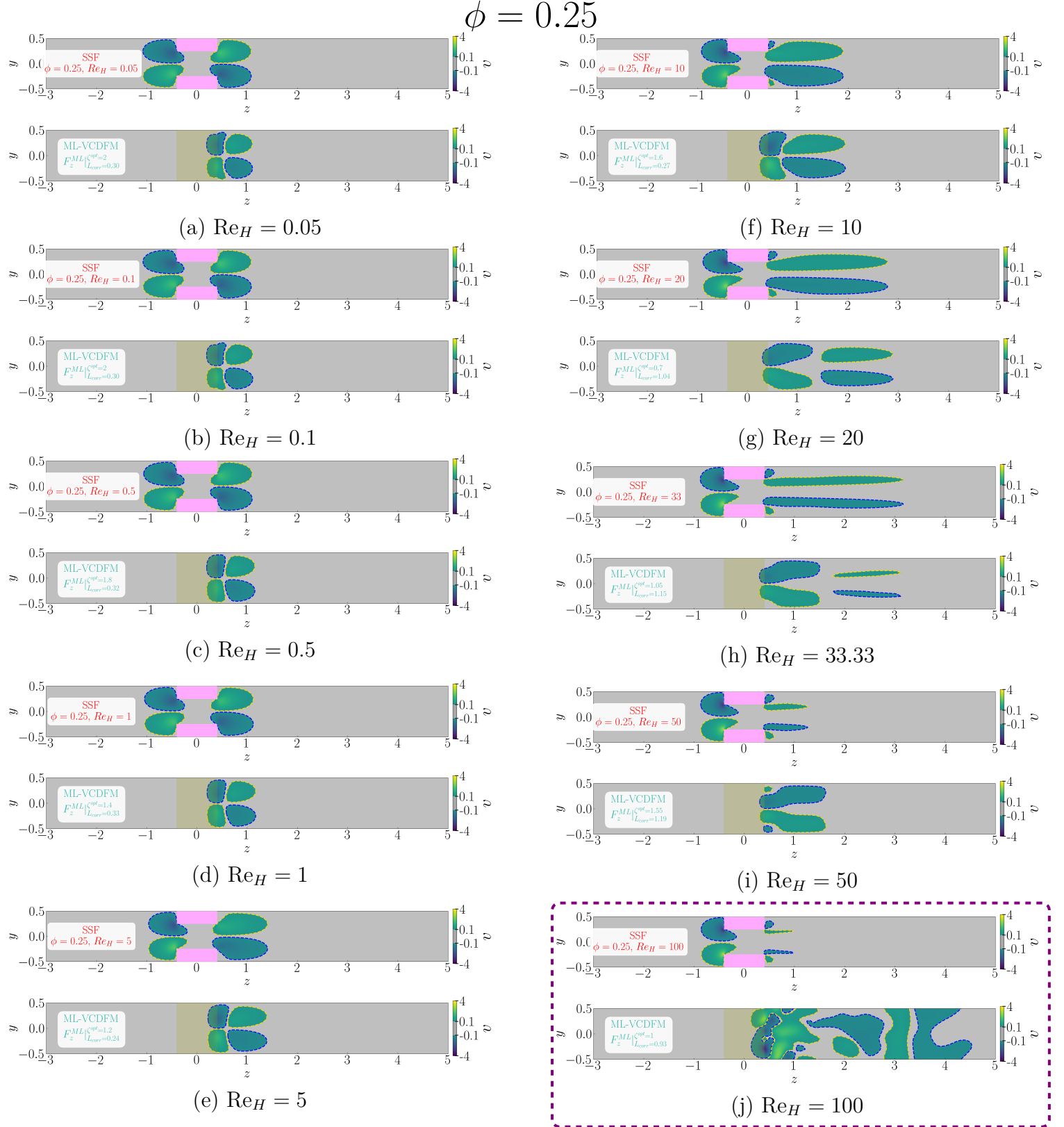


Figure D.14: Transverse velocity component v in the streamwise (z, y) -plane for the SRG at $\phi = 0.25$ and $\alpha = 0^\circ$. For each Reynolds number, the upper panel shows the geometry-resolved small-scale flow (SSF), while the lower panel shows the ML-VCDFM reconstruction using F_z^{ML} . The violet dashed rectangle marks the extrapolation case $Re_H = 100$, where significant differences from the SSF place the ML-VCDFM reconstruction outside its effective validity range.

D.2 Detailed Validation Cases Within the Effective Regime

This section provides detailed comparisons between the geometry-resolved small-scale flow (SSF) and the VCDFM for the configurations that remain within the effective validity regime identified above. For each valid case, the comparison includes: (i) cross-sectional L_2 -norms and centerline quantities, (ii) downstream flow organization in the streamwise (z, y) -plane, (iii) transverse velocity profiles at selected downstream locations, and (iv) transverse (x, y) -plane distributions.

For all validation cases, the geometry-resolved small-scale flow (SSF), the Darcy–Forchheimer model (DFM), and the velocity-corrected Darcy–Forchheimer model (VCDFM) are compared. When applicable, three VCDFM variants are considered: SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM.

In all figures, the SSF solution is shown in red, the DFM solution in olive, SSF-VCDFM in blue, TIDFT-VCDFM in magenta, and ML-VCDFM in cyan. The blue shaded region denotes the physical grid location in the SSF ($z = \pm\Gamma/2$). When shown, $\mathcal{L}_{\text{entrance}}$ and $\mathcal{L}_{\text{exit}}$ correspond to the entrance and exit regions identified from the SSF, whereas $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ denote the forcing-response regions used in the VCDFM analysis.

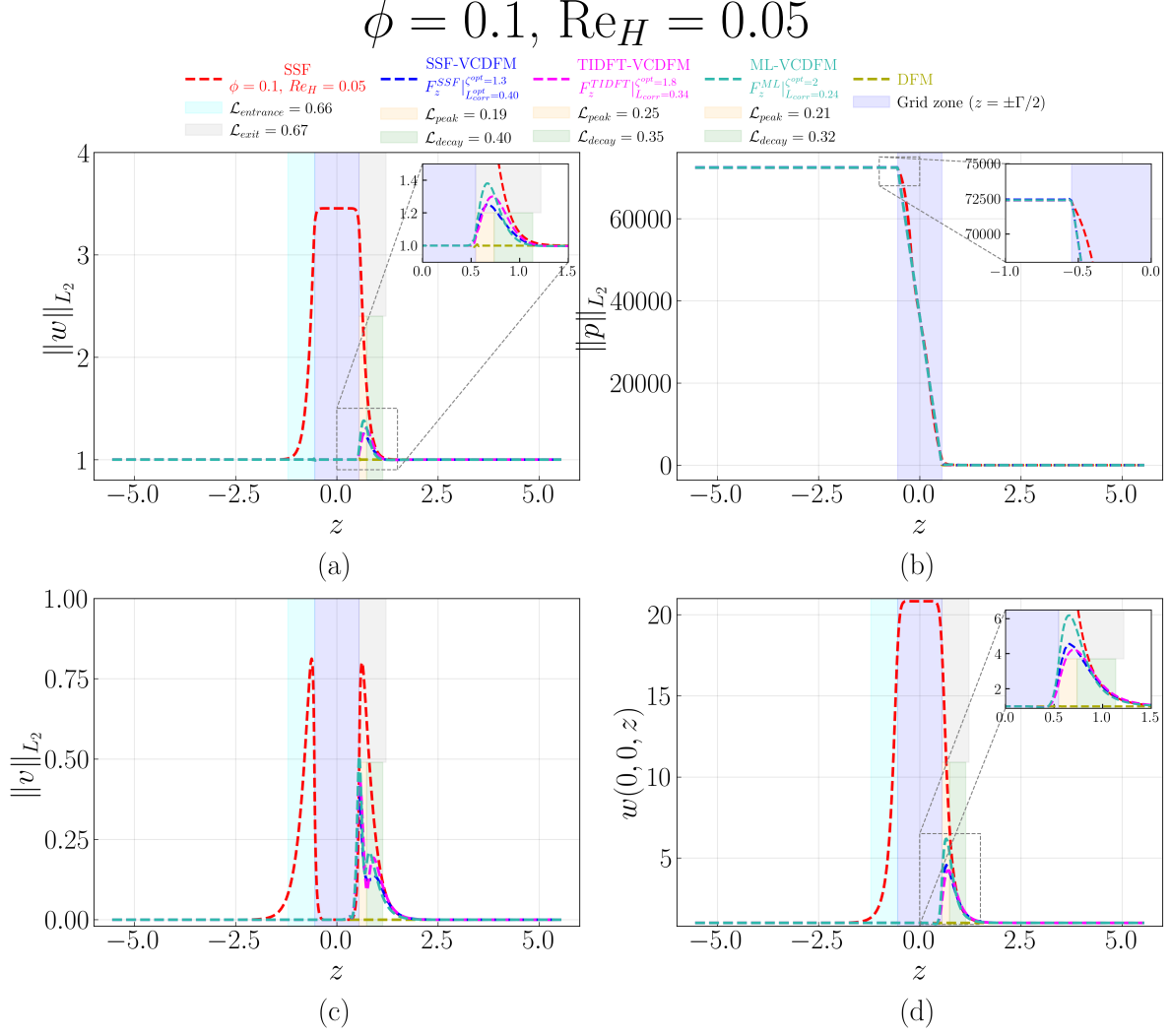


Figure D.15: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.05$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

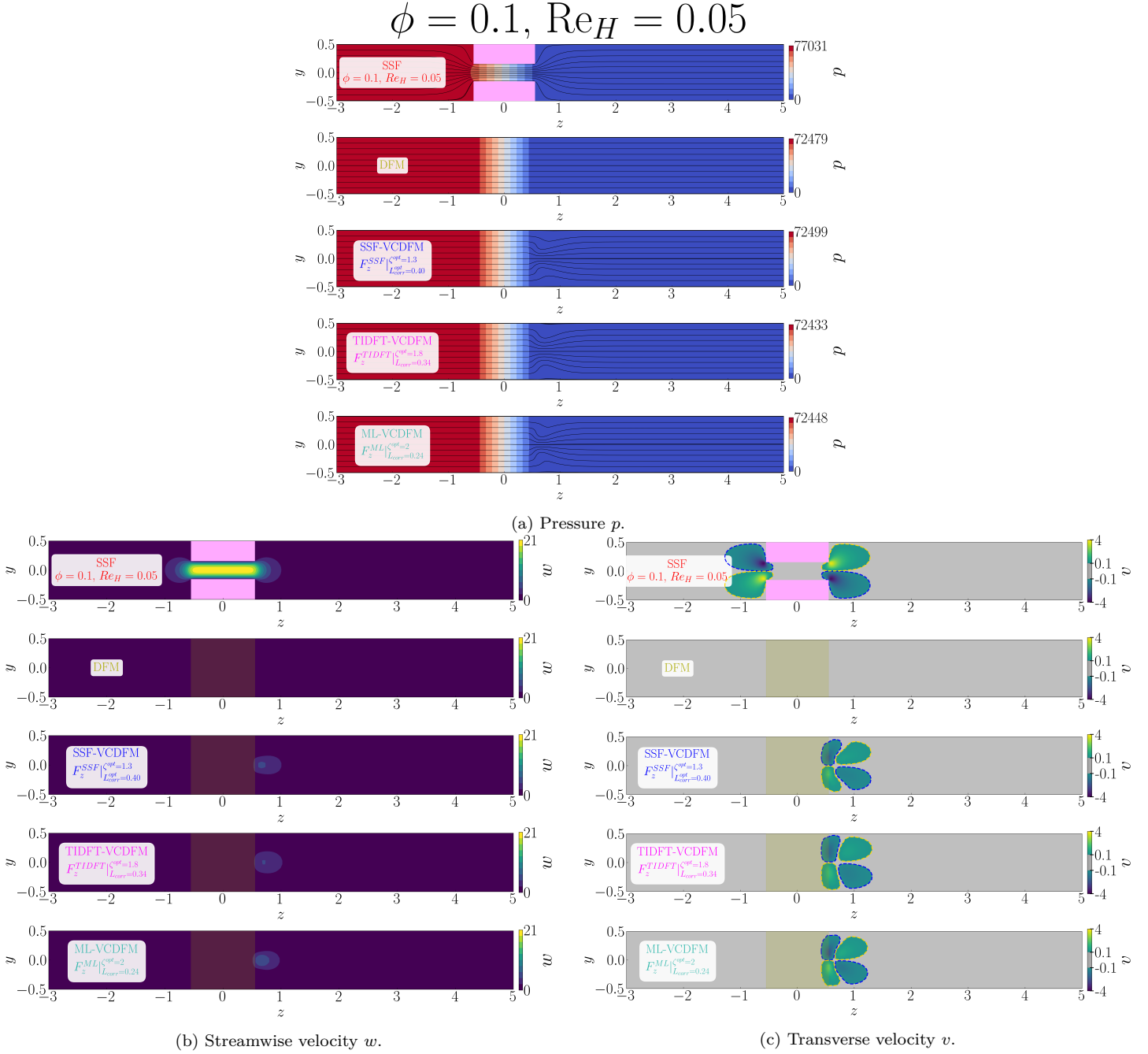


Figure D.16: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $\text{Re}_H = 0.05$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

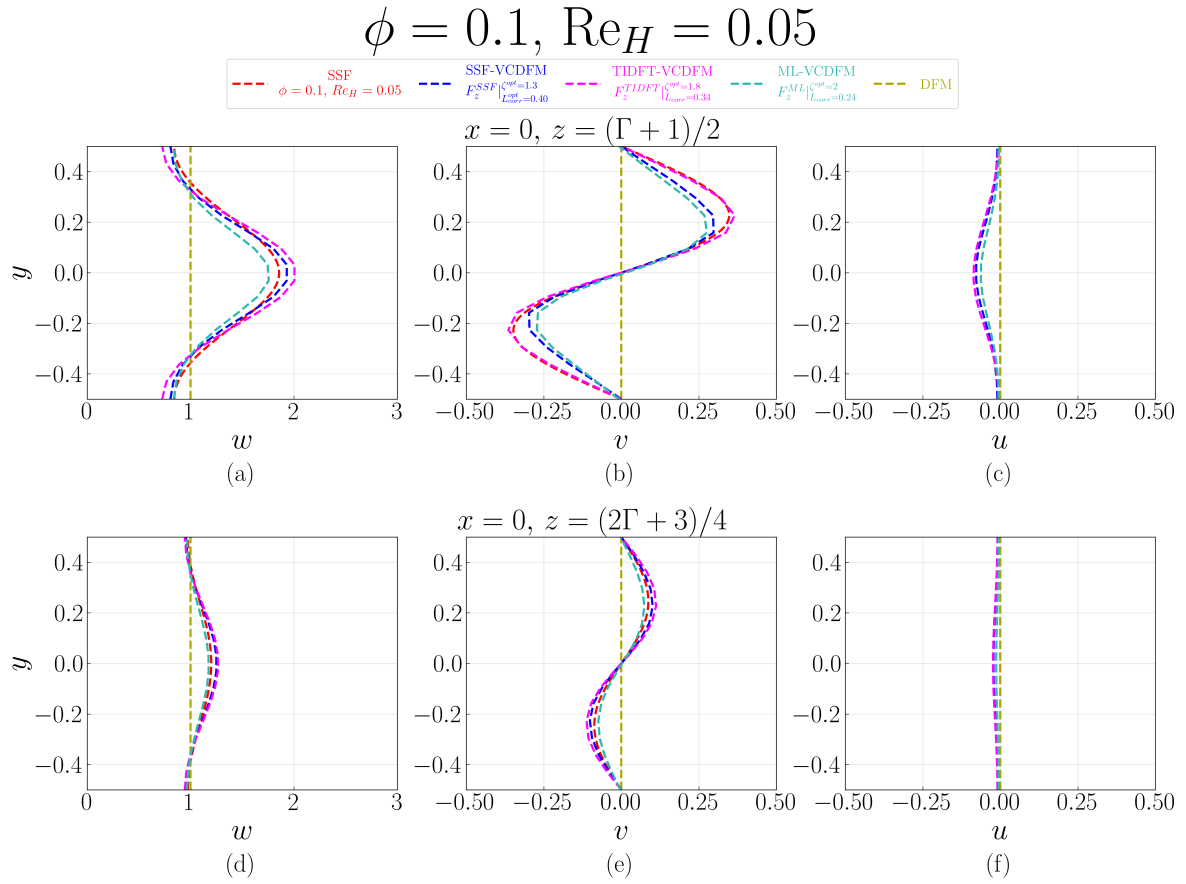


Figure D.17: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.05$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 1)/2$ in (a–c) and $z = (2\Gamma + 3)/4$ in (d–f).

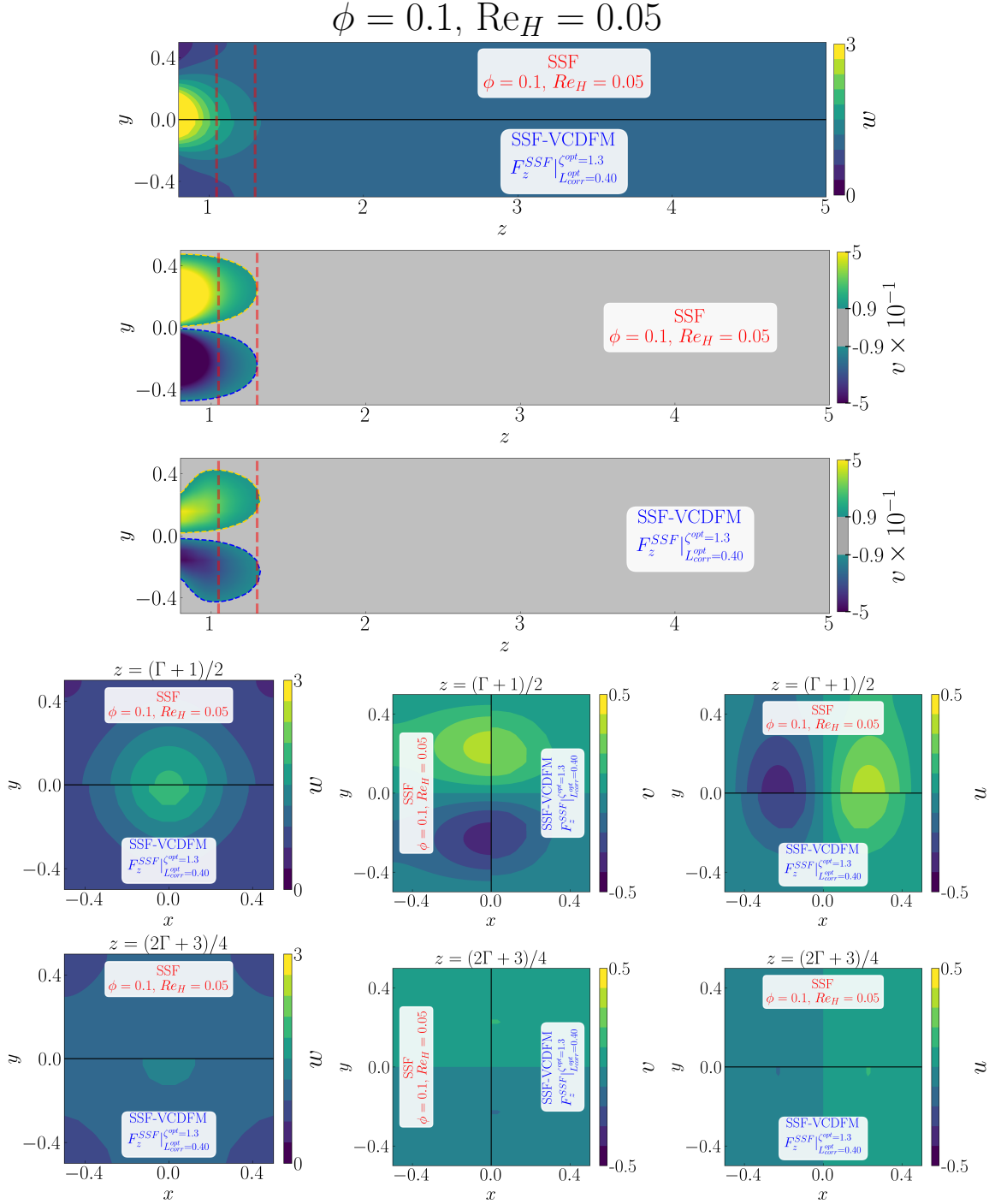


Figure D.18: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.05$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

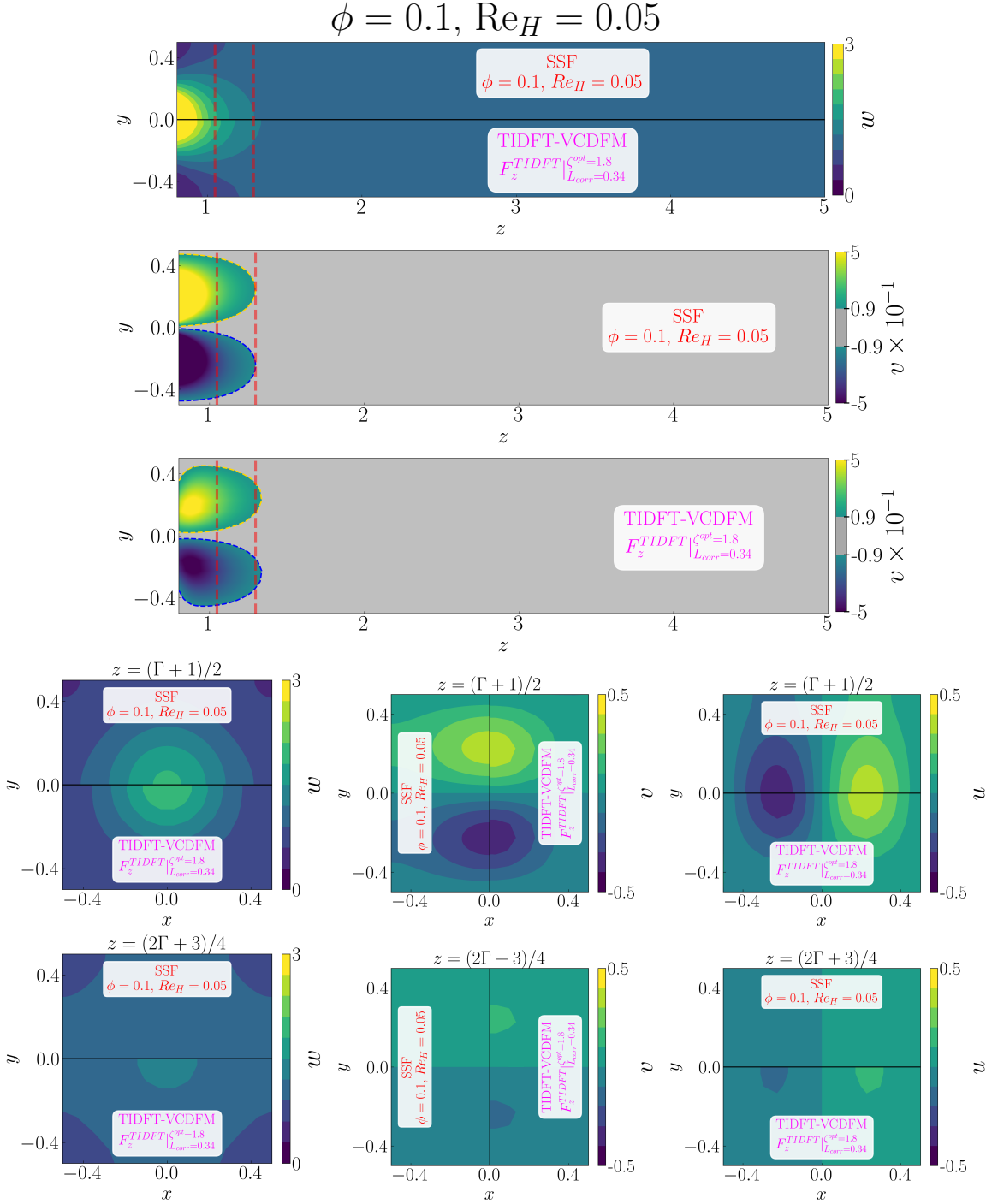


Figure D.19: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.05$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

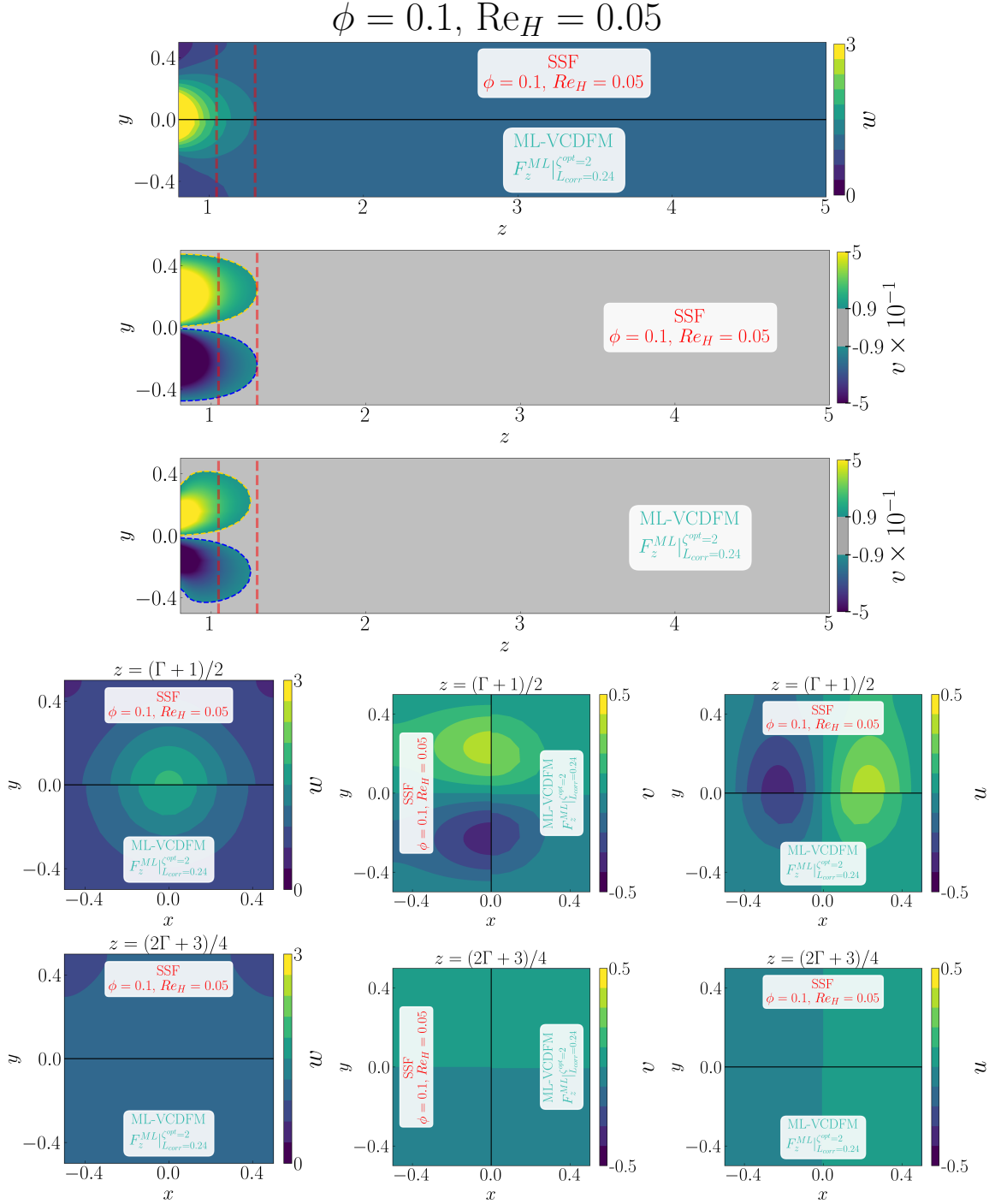


Figure D.20: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.05$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

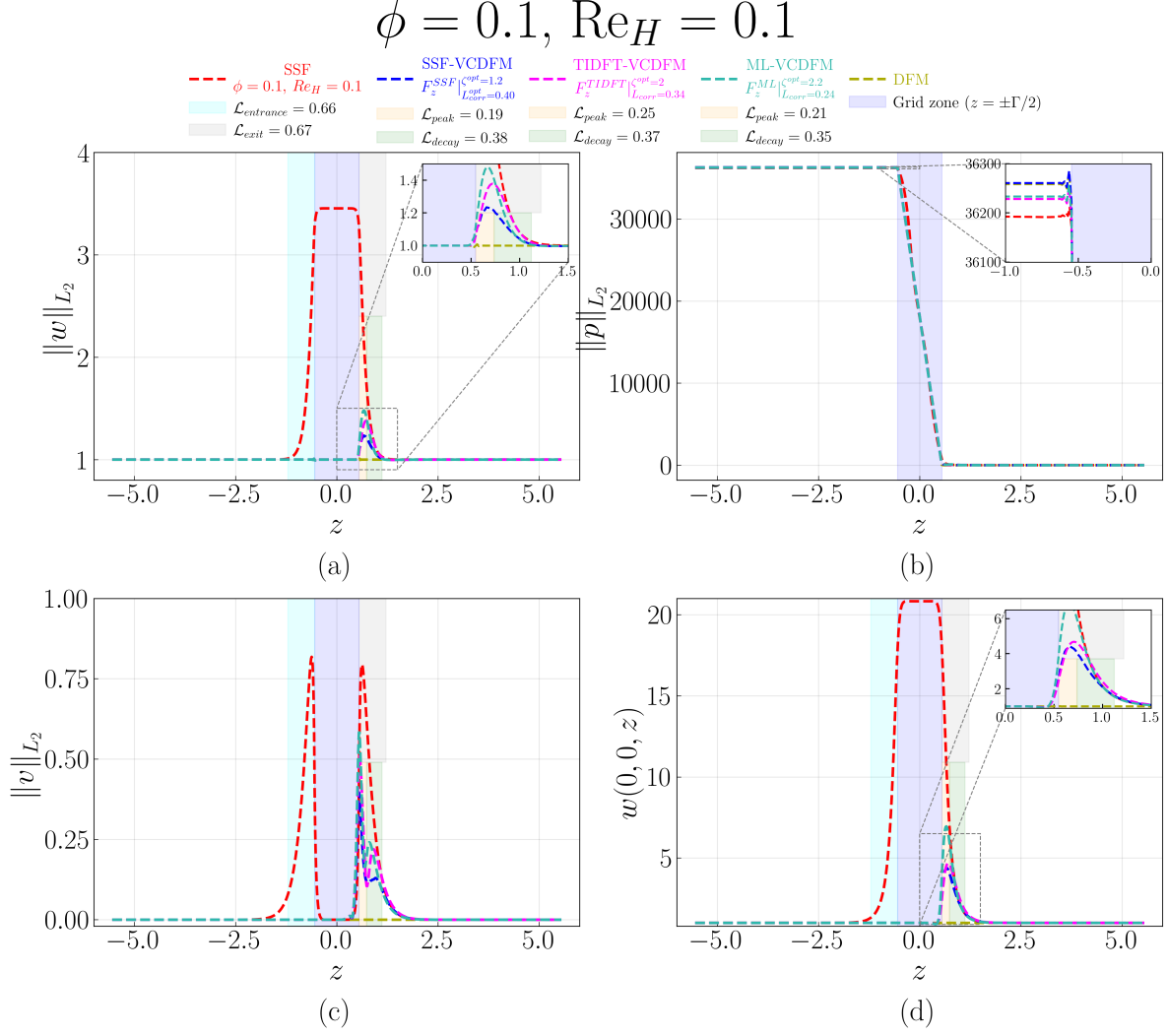


Figure D.21: Streamwise evolution of (a) $\|w\|_{L_2}$, (b) $\|p\|_{L_2}$, (c) $\|v\|_{L_2}$, and (d) centerline velocity $w(0,0,z)$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.1$. The figure compares the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Shading marks the grid and the characteristic entrance, exit, peak, and decay regions. Only the SSF-based $\mathcal{L}_{\text{peak}}$ and $\mathcal{L}_{\text{decay}}$ are marked on the plot; the corresponding TIDFT-VCDFM and ML-VCDFM values appear in the legend.

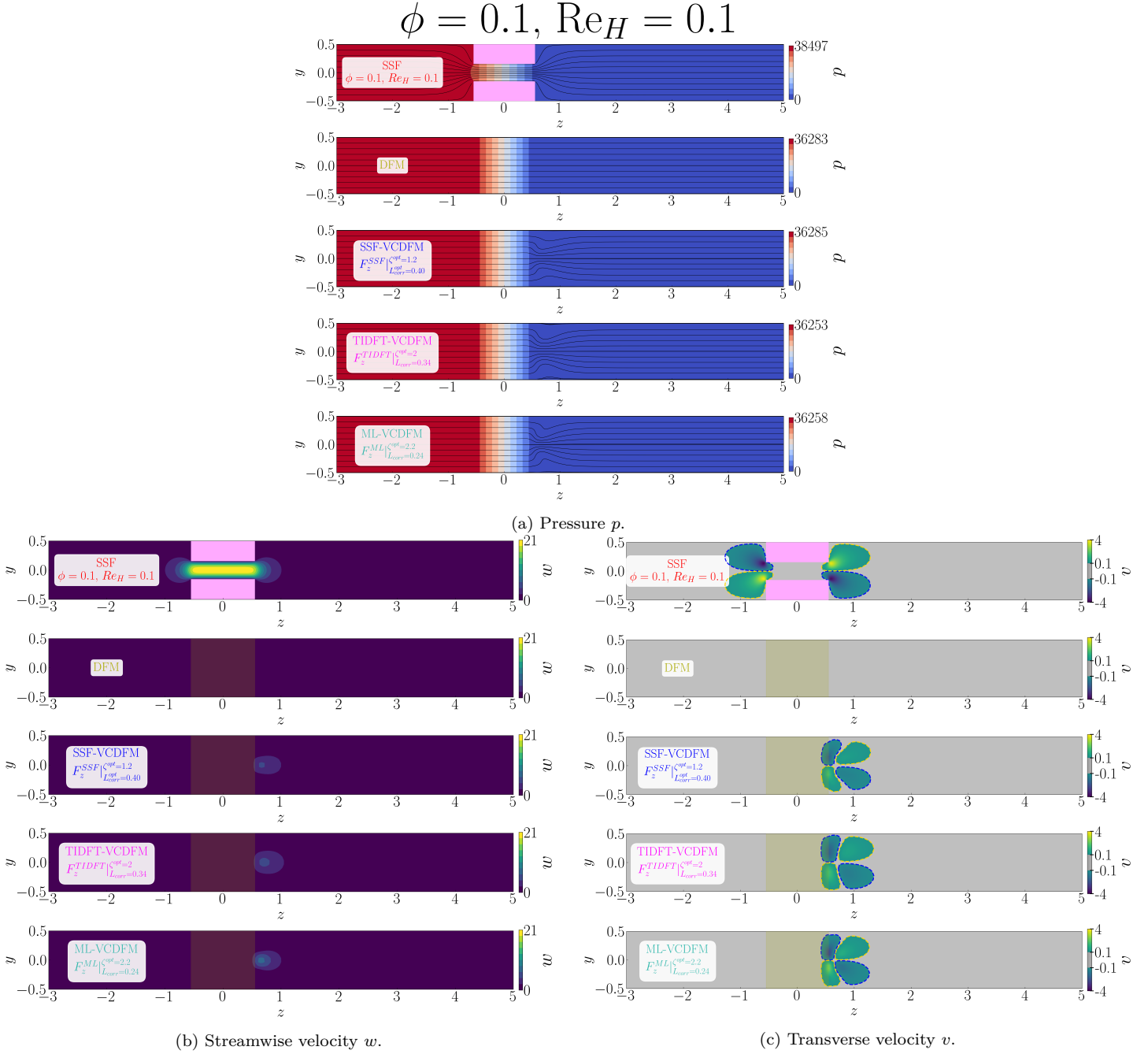


Figure D.22: Flow organization in the streamwise (z, y) -plane for the 1×1 SRG at $\phi = 0.1$, $Re_H = 0.1$, and $\alpha = 0^\circ$. The blocks show (a) pressure p with streamlines, (b) streamwise velocity w , and (c) transverse velocity v . Within each block, the rows show, from top to bottom, the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Pink shading marks the grid walls in the SSF; olive shading marks the equivalent porous zone in the model solutions.

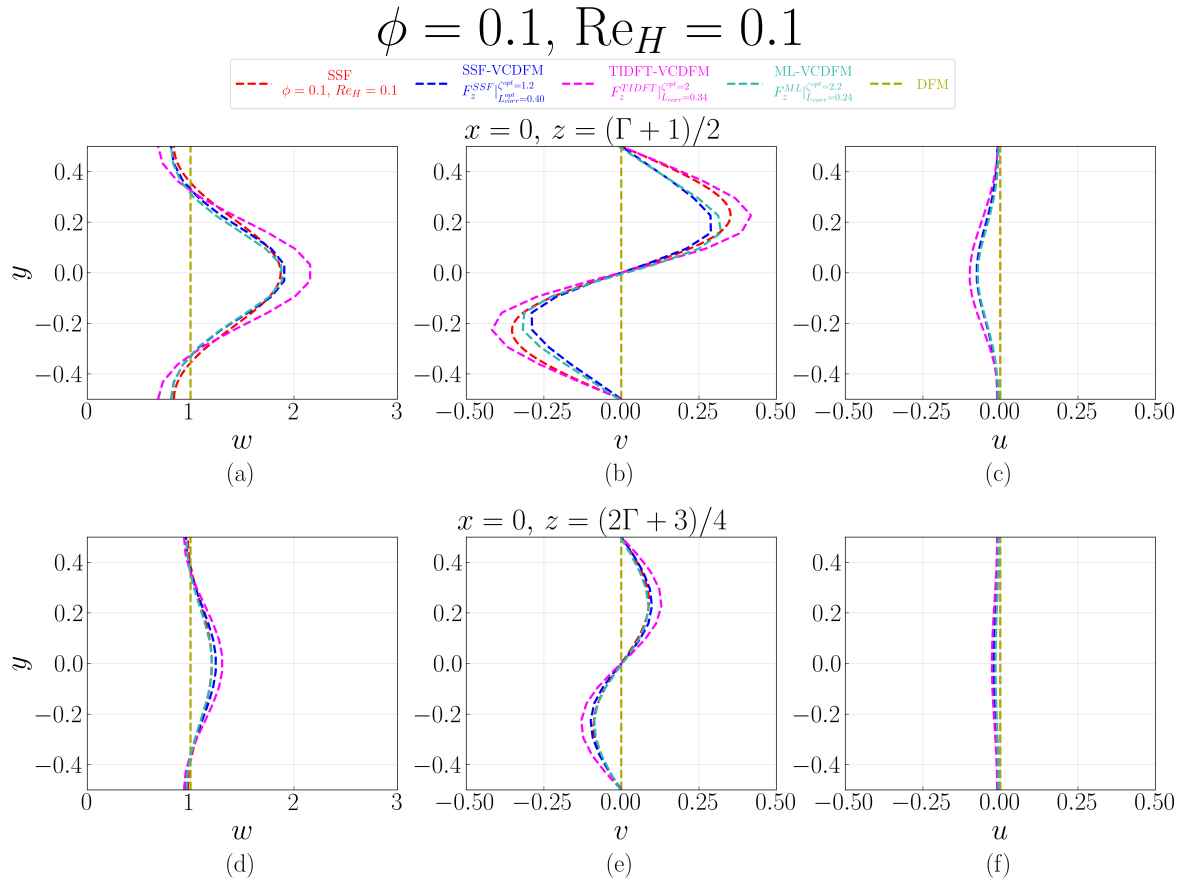


Figure D.23: Velocity profiles at $x = 0$ for the 1×1 SRG at $\phi = 0.1$ and $\text{Re}_H = 0.1$, comparing the SSF, DFM, SSF-VCDFM, TIDFT-VCDFM, and ML-VCDFM. Profiles of $w(y)$, $v(y)$, and $u(y)$ are shown at $z = (\Gamma + 1)/2$ in (a–c) and $z = (2\Gamma + 3)/4$ in (d–f).

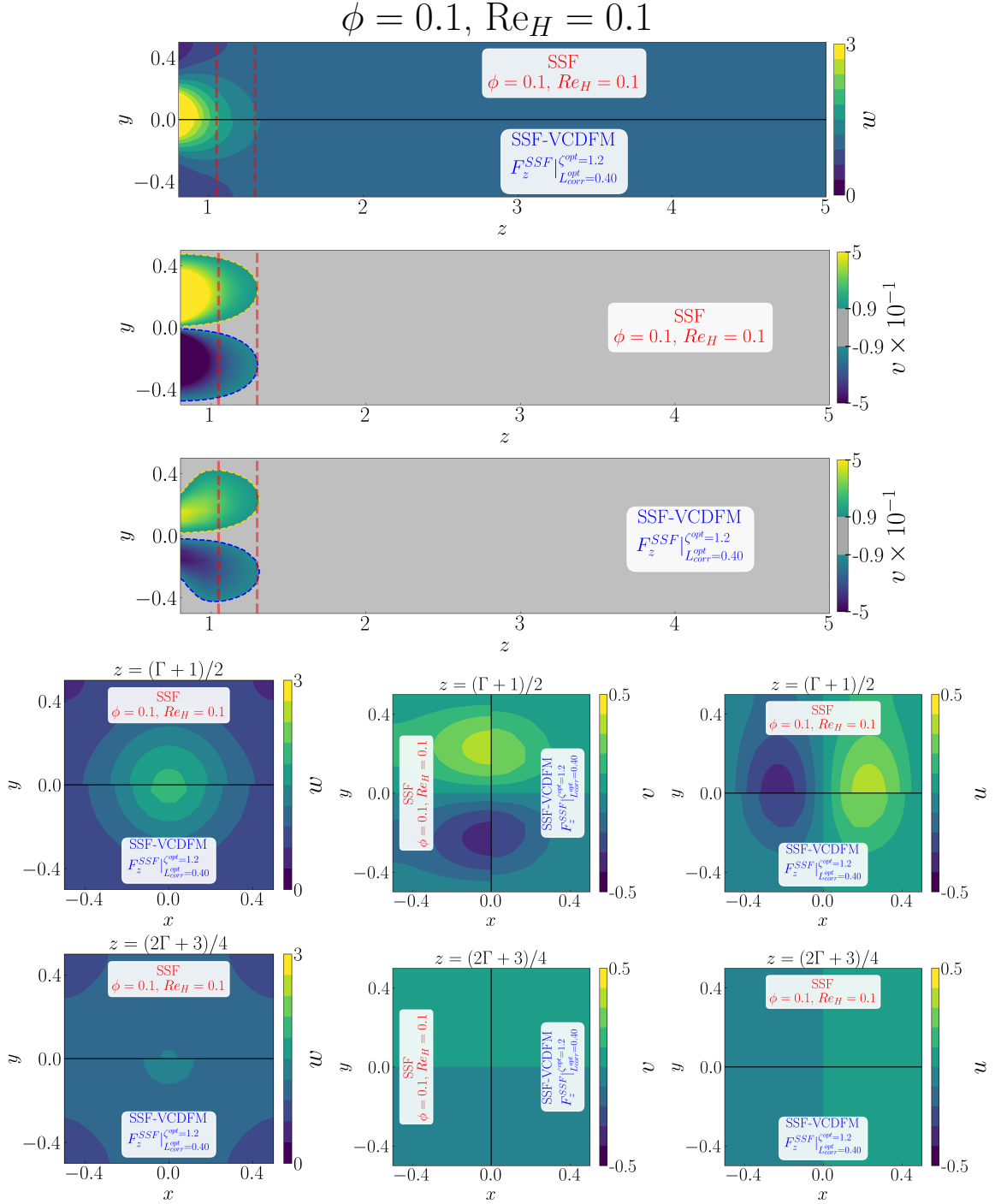


Figure D.24: Detailed comparison between the geometry-resolved small-scale flow (SSF) and SSF-VCDFM using F_z^{SSF} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.1$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

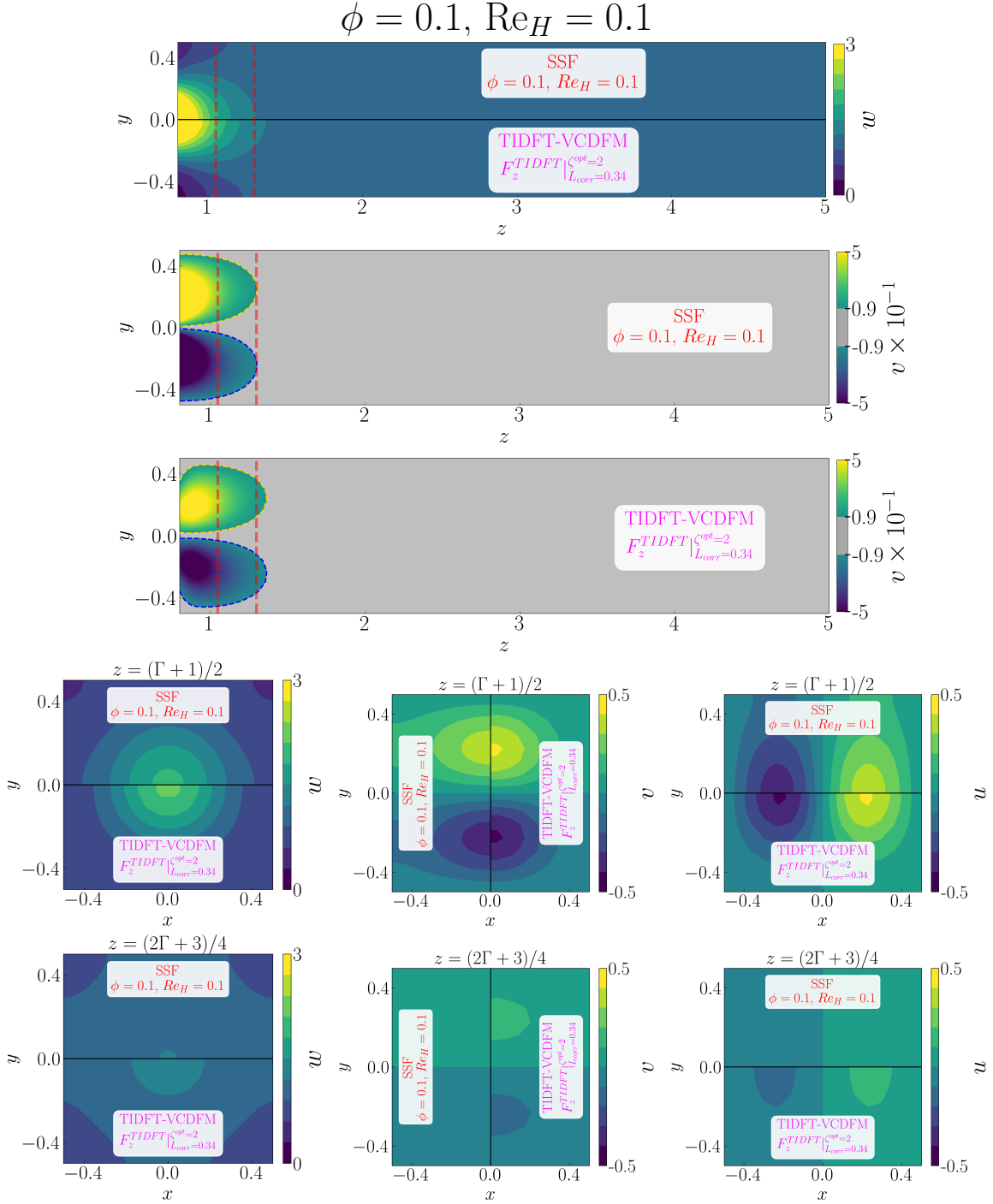


Figure D.25: Detailed comparison between the geometry-resolved small-scale flow (SSF) and TIDFT-VCDFM using F_z^{TIDFT} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.1$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.

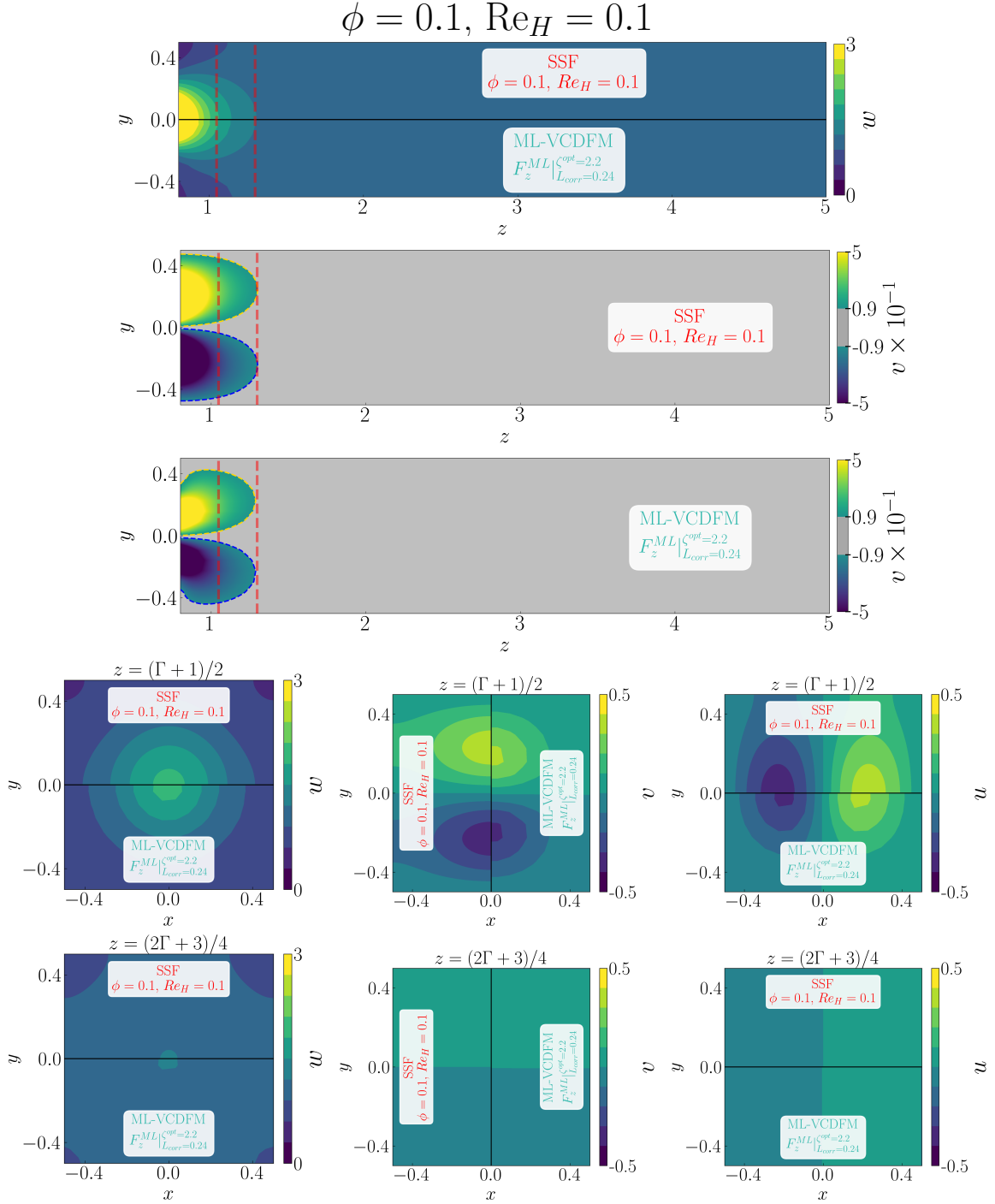


Figure D.26: Detailed comparison between the geometry-resolved small-scale flow (SSF) and ML-VCDFM using F_z^{ML} for the 1×1 SRG at $\phi = 0.1$ and $Re_H = 0.1$. The upper panels compare the streamwise velocity w and transverse velocity v in the central (z, y) -plane, restricted to the correction region. The lower panels compare the corresponding transverse (x, y) -plane distributions of w , v , and u , from left to right, at $z = (\Gamma + 1)/2$ in the first row and $z = (2\Gamma + 3)/4$ in the second row.